

LM

$$Y_i = \underbrace{\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}}_{\mathbf{x}_i^\top \boldsymbol{\beta}} + \varepsilon_i; \quad i = 1, \dots, n$$

$$\mathbf{x}_i^\top = (1, x_{i1}, \dots, x_{ip})$$

Assumptions:

$$\mathbb{E}(\varepsilon_i) = 0$$

$$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$$

$$\text{Var}(\varepsilon_i) = \sigma^2$$

$$\{\varepsilon_i \mid i = 1, \dots, n\} \text{ indep.}$$

$$\text{from (2.3), (2.4) follows } \text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$$

$$\varepsilon_i \sim N(0, \sigma^2) \text{ and (2.4)}$$

$$\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$$

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

$$\Rightarrow \mathbb{E}(Y|\mathbf{X}) = \mathbf{X}^\top \boldsymbol{\beta}$$

Prediction: $\hat{Y}_{n+1} = \mathbf{x}_{n+1}^\top \hat{\boldsymbol{\beta}}$

OLS - Estimator: $\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \underbrace{(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})}_{\sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2} \Rightarrow \hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}. \quad \hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$

Wald Test

Consider the quadratic form corresponding to the linear Hypothesis
 $H_0 : \mathbf{A}\boldsymbol{\beta} = \mathbf{c}$:

$$(\mathbf{A}\hat{\boldsymbol{\beta}} - \mathbf{c})^\top (\mathbf{A}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{A}^\top)^{-1} (\mathbf{A}\hat{\boldsymbol{\beta}} - \mathbf{c})$$

$$\text{Var}(\mathbf{A}\hat{\boldsymbol{\beta}} - \mathbf{c}) = \sigma^2 \mathbf{A}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{A}^\top$$

$$SSH := (\mathbf{A}\hat{\boldsymbol{\beta}} - \mathbf{c})^\top (\mathbf{A}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{A}^\top)^{-1} (\mathbf{A}\hat{\boldsymbol{\beta}} - \mathbf{c})$$

$$\sigma^{-2} SSH \sim \chi^2(a, \sigma^{-2} (\mathbf{A}\boldsymbol{\beta} - \mathbf{c})^\top (\mathbf{A}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{A}^\top)^{-1} (\mathbf{A}\boldsymbol{\beta} - \mathbf{c}))$$

$$MSH := \frac{SSH}{a}$$

$$\frac{MSH}{MSE} \sim F(a, n - p', \sigma^{-2} (\mathbf{A}\boldsymbol{\beta} - \mathbf{c})^\top (\mathbf{A}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{A}^\top)^{-1} (\mathbf{A}\boldsymbol{\beta} - \mathbf{c}))$$

SSH: Sum of squares describing the deviation from the hypothesis
 $\mathbf{A}\boldsymbol{\beta} = \mathbf{c}$.

Wald Test: $H_0 : \mathbf{A}\boldsymbol{\beta} = \mathbf{c}$. Reject H_0 , if:

$$TF = MSH/MSE > F_{1-\alpha}(a, n - p') \quad (2.13)$$

Likelihood Ratio Test

- Take H_1 if $\frac{\mathcal{L}(Y, X, \theta_1)}{\mathcal{L}(Y, X, \theta_0)} > k$

R²

$$\underbrace{(\mathbf{Y} - \bar{\mathbf{Y}})^\top (\mathbf{Y} - \bar{\mathbf{Y}})}_{SST} = \underbrace{(\mathbf{Y} - \hat{\mathbf{Y}})^\top (\mathbf{Y} - \hat{\mathbf{Y}})}_{SSE} + \underbrace{(\hat{\mathbf{Y}} - \bar{\mathbf{Y}})^\top (\hat{\mathbf{Y}} - \bar{\mathbf{Y}})}_{SSM} \quad (2.16)$$

Interpretation:

SST : dispersion of the observed variables around the mean

SSE : error difference between observed and the predicted value

SSM : measure that describes how well our line fits the data

Coefficient of determination:

$$R^2 = SSM/SST \quad \text{How much variance in the data are we able to explain}$$

Interactions

- Discrete variables: use all products of dummy variables
- continuous variables: product of the variables
- discrete and continuous: use all products

$$Y_{kl} = \beta_0 + \tau_k + \beta_3 X_{kl} + \beta_k X_{kl} + \varepsilon_{kl} \quad (k = 1, 2)$$

$$Y_{kl} = \beta_0 + \beta_3 X_{kl} + \varepsilon_{kl} \quad (k = 3)$$

matrix representation (3 groups 2 observations per group)

$$\mathbf{X} = \begin{pmatrix} 1 & 1 & 0 & x_1 & x_1 & 0 \\ 1 & 1 & 0 & x_2 & x_2 & 0 \\ 1 & 0 & 1 & x_3 & 0 & x_3 \\ 1 & 0 & 1 & x_4 & 0 & x_4 \\ 1 & 0 & 0 & x_5 & 0 & 0 \\ 1 & 0 & 0 & x_6 & 0 & 0 \end{pmatrix}, \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \tau_1 \\ \tau_2 \\ \beta_3 \\ \beta_1 \\ \beta_2 \end{pmatrix} \begin{array}{l} \text{- intercept} \\ \text{- dummy} \\ \text{- dummy} \\ \text{- numerical} \\ \text{- interaction} \\ \text{- interaction} \end{array}$$

GLM $\Rightarrow E(Y_i | x_i) = h(x_i^T \beta)$ h : response function

Structure: - no residuals but distributional assumptions

$$\mu_i = E(Y_i | x_i) = h(x_i^T \beta)$$

$\Rightarrow$ Exp-fam:

$$f(y_i | \theta_i) = \exp \left(\frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right),$$

with

- θ natural or canonical parameter, $(\theta = g(\beta) = x^T \beta \Rightarrow \text{canonical link})$
- $b(\theta)$ function (normalization),
- ϕ scales or dispersion parameters,
- $c(y_i, \phi)$ function

$$\Rightarrow \begin{aligned} E(Y_i) &= b'(\theta_i) \\ \text{Var}(Y_i) &= \phi b''(\theta_i). \end{aligned}$$

$$\Rightarrow \text{Gaussian: } f(y) = \exp \left\{ -\frac{1}{2} \left(\frac{y - \mu}{\sigma} \right)^2 - \log(\sqrt{2\pi} \sigma) \right\} = \exp \left\{ \frac{y\mu - \mu^2/2}{\sigma^2} - \frac{y^2}{2\sigma^2} - \log(\sqrt{2\pi} \sigma) \right\}.$$

$$\Rightarrow \text{Gamma: } f(y) = \frac{1}{\Gamma(\nu)} \left(\frac{\nu}{\mu} \right)^\nu y^{\nu-1} \exp \left(-\frac{\nu}{\mu} y \right) \\ = \exp \left(\underbrace{-\frac{(1/\mu)y - \log(\mu)}{1/\nu}}_{\theta} + \nu \log(\nu) + (\nu - 1) \log(y) - \log(\Gamma(\nu)) \right).$$

Exponential family parametrization

Dispersion parameter $\phi = 1/\nu$ and $\theta(\mu) = -1/\mu$, $b(\theta) = -\log(-\theta)$.

Natural link function:

$$g(\mu) = -1/\mu$$

Score function for Expfam: $\left(\frac{\partial \ell(\beta)}{\partial \beta} \right)$

$$s(\beta) = \sum_{i=1}^n x_i \frac{h'(\eta_i)}{\sigma_i^2} (y_i - \mu_i) \\ = \underbrace{\mathbf{X}^T}_{\text{data}} \underbrace{\mathbf{D}}_{\text{weights}} \underbrace{\Sigma^{-1}}_{\text{residual}} (\mathbf{y} - \boldsymbol{\mu}),$$

$$\mathbf{D} = \text{diag}(h'(\eta_1), \dots, h'(\eta_n))$$

$$\sigma_i^2 = \text{Var}(y_i | \theta_i) = \phi b''(\theta_i)$$

$$\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$$

$$\eta_i = x_i^T \beta$$

$$\mu_i = E(y_i | \theta_i)$$

$$\boldsymbol{\mu} = (\mu_1, \dots, \mu_n)^T$$

with not link: $s(\beta) = \sum_{i=1}^n x_i (y_i - \mu_i) / \phi$

Expected Fisher information

$$\mathbf{F}(\beta) = E \left(-\frac{\partial^2 \ell(\beta)}{\partial \beta \partial \beta^T} \right) = \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^T \omega_i \quad \omega_i = \frac{(h'(\eta_i))^2}{\sigma_i^2}$$

Asymptotic properties of ML

$$\hat{\beta} \overset{a}{\sim} N(\beta, \mathbf{F}^{-1}(\hat{\beta}))$$

ML: LR-Test $H_0: \mathbf{A}\beta = \mathbf{c}$ vs. $H_1: \mathbf{A}\beta \neq \mathbf{c} \Rightarrow lr = -2(\ell(\hat{\beta}) - \ell(\hat{\beta})) \overset{a}{\sim} \chi^2(\text{rank}(\mathbf{A}))$

Deviance Let $\ell(\mathbf{y}; \hat{\boldsymbol{\mu}}, \phi)$ be the maximized log-likelihood with $\mathbf{y}^T = (y_1, \dots, y_n)$ and $\hat{\boldsymbol{\mu}}^T = (\hat{\mu}_1, \dots, \hat{\mu}_n)$, $\hat{\mu}_i = h(x_i^T \hat{\beta})$.

The deviance is defined as

$$D(\mathbf{y}, \hat{\boldsymbol{\mu}}) = -\phi 2 \{ \ell(\mathbf{y}; \hat{\boldsymbol{\mu}}, \phi) - \ell(\mathbf{y}; \mathbf{y}, \phi) \} \\ = 2 \sum_{i=1}^n \{ y_i (\theta(y_i) - \theta(\hat{\mu}_i)) - (b(\theta(y_i)) - b(\theta(\hat{\mu}_i))) \}.$$

pretending that the mean that is estimated is exactly the observed data

GLM 2

Quasi Likelihood: no distr. assumption

Basic idea: specify expected value and variance

$$\mathbb{E}(Y_i | \mathbf{x}_i) = \mu_i = h(\mathbf{x}_i^\top \beta), \quad \text{Var}((Y_i | \mathbf{x}_i) = \sigma_i^2(\mu_i) = \phi v(\mu_i), \quad (2.29)$$

Define

$$\begin{aligned} \mathbf{s}_Q(\beta) &= \mathbf{X}^\top \mathbf{D}(\beta) \Sigma^{-1}(\beta) (\mathbf{y} - \boldsymbol{\mu}) \\ &= \sum_{i=1}^n \mathbf{x}_i \frac{h'(\mathbf{x}_i^\top \beta)}{\phi v(\mu_i)} (y_i - h(\mathbf{x}_i^\top \beta)) \end{aligned} \quad (2.30)$$

=> we have no Likelihood or score function !!

Logistic Regression Model:

$$\mathbb{E}(Y_i | \mathbf{x}_i) = \mathbb{P}(Y_i = 1 | \mathbf{x}_i) = \frac{\exp(\mathbf{x}_i^\top \beta)}{1 + \exp(\mathbf{x}_i^\top \beta)}$$

- The **odds** $\pi_i / (1 - \pi_i) = \mathbb{P}(Y_i = 1 | \mathbf{x}_i) / \mathbb{P}(Y_i = 0 | \mathbf{x}_i)$ follow the multiplicative model

$$\frac{\mathbb{P}(Y_i = 1 | \mathbf{x}_i)}{\mathbb{P}(Y_i = 0 | \mathbf{x}_i)} = \exp(\beta_0) \cdot \exp(x_{i1}\beta_1) \cdot \dots \cdot \exp(x_{ip}\beta_p) .$$

Response is log-odds:

$$g(\pi_i) = \log \left(\frac{\pi_i}{1 - \pi_i} \right)$$

- $\beta_1 > 0 \Rightarrow \exp(\beta_1) > 1$: odds increase with growing x_1
- $\beta_1 < 0 \Rightarrow \exp(\beta_1) < 1$: odds decrease with growing x_1
- $\beta_1 = 0 \Rightarrow \exp(\beta_1) = 1$: odds remain unchanged.

Log-Linear Poisson Model

$$\mathbb{E}(Y_i | \mathbf{x}_i) = \lambda_i = \exp(\eta_i) = \exp(\beta_0) \exp(\beta_1 x_{i1}) \cdot \dots \cdot \exp(\beta_p x_{ip})$$

$$\log(\lambda_i) = \eta_i = \mathbf{x}_i^\top \beta = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} .$$

Poisson with Offset

$$\begin{aligned} \frac{\mathbb{E}(Y_i | \mathbf{x})}{N_i} &= \exp(\mathbf{x}_i^\top \beta) \\ Y_i &\sim \text{Po}(N_i \cdot \exp(\mathbf{x}_i^\top \beta)) \end{aligned} \quad \Rightarrow \quad \mathbb{E}(Y_i | \mathbf{x}) = \exp(\mathbf{x}_i^\top \beta + \log(N_i))$$

Negative Binomial: more flexible than Poisson distr.

$$f(y | \mu, \phi) = \binom{y + \phi - 1}{y} \left(\frac{\mu}{\mu + \phi} \right)^y \left(\frac{\phi}{\mu + \phi} \right)^\phi$$

$$\mathbb{E}(Y) = \mu$$

$$\text{Var}(Y) = \mu + \frac{\mu^2}{\phi}$$

Beta Regression

- Appropriate if $Y \in (0, 1)$, e.g., rates, proportions, etc.
- Beta distribution: Continuous distribution for $0 < y < 1$ with two shape parameters $a, b > 0$
 $\Rightarrow$ interpretation not straightforward
- Alternatively: Parametrize model in the distribution's mean $\mu = a/(a + b)$ and precision $\tau = a + b$, i.e.,

$$Y_i | \mathbf{x}_i \sim \text{Be}(\mu(\mathbf{x}_i), \tau(\mathbf{x}_i))$$

- $\text{Var}(Y) = \frac{\mu(1-\mu)}{1+\tau}$ — additional flexibility by the τ

GLM3 - Familien & Interpretation

Family	natural Link	Interpretation für $x \rightarrow x+1$
gaussian	identity $g(\mu) = \mu$	Erwarteter Wert erhöht sich um β
poisson	$g(\mu) = \log(\mu)$	Expected log-value increases by β ↳ or: expected value multiplied with $\exp(\beta)$
binomial	Logit $g(p) = \log\left(\frac{p}{1-p}\right)$	expected log-odds increase by β ↳ or: odds-ratio multiplied with $\exp(\beta)$
gamma	$g(\mu) = 1/\mu$	inverse of expected value increases by β
inverse-gamma	$g(\mu) = 1/\mu^2$	inverse of squared expected value increases by β
	other Link-functions	
	probit: $g(\mu) = \Phi^{-1}(\mu)$	
	square root: $g(\mu) = \sqrt{\mu}$	expected value of square root increases by β

$\Rightarrow$ „keeping all other variables constant“

Linear Mixed Models

- grouped data & longitudinal data $\Rightarrow$ no independence

$\Rightarrow$ Assumption: if we remove individual intercepts and slopes (and maybe sth else)

then we have independent data

Random intercept

conditional

$\hookrightarrow$ quasi model with fixed effects for each individual

$$\begin{aligned} Y_{ij} &= \mathbf{x}_{ij}^\top \boldsymbol{\beta} + \gamma_i + \epsilon_{ij} \quad i = 1, \dots, g; \quad j = 1, \dots, n_i \\ \epsilon &\sim N(\mathbf{0}, \sigma^2 \mathbf{I}) \\ \gamma_i &\sim N(0, \sigma_\gamma^2) \end{aligned}$$

γ_i and ϵ independent

individual intercept deviation
it's a random variable
 β_0 is the overall intercept

marginal

$\hookrightarrow$ don't care for individuals
 $\hookrightarrow$ want global effect and account for variance

$$\begin{aligned} Y_{ij} &= \mathbf{x}_{ij}^\top \boldsymbol{\beta} + \delta_{ij} \\ \delta_{ij} &= \epsilon_{ij} + \gamma_i \\ \text{Var}(\delta_{ij}) &= \sigma^2 + \sigma_\gamma^2; \quad E(\delta_{ij}) = 0 \\ \text{COV}(\delta_{i_1 j_1}, \delta_{i_2 j_2}) &= \sigma_\gamma^2 \Rightarrow \text{same subject} \\ \text{COV}(\delta_{i_1 j_1}, \delta_{i_2 j_2}) &= 0 \text{ for } i_1 \neq i_2 \Rightarrow \text{different subjects} \end{aligned}$$

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\delta}$$

$$\boldsymbol{\delta} \sim N(\mathbf{0}, \sigma^2 \mathbf{I} + \text{diag}[\sigma_\gamma^2 \mathbf{e}_i \mathbf{e}_i^\top])$$

$\mathbf{e}_i$:= 1-vector of length n_i

$$\boldsymbol{\delta} = \begin{pmatrix} \delta_{11} \\ \delta_{12} \\ \delta_{21} \\ \delta_{22} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}; \begin{pmatrix} \sigma^2 + \sigma_\gamma^2 & \sigma_\gamma^2 & 0 & 0 \\ \sigma_\gamma^2 & \sigma^2 + \sigma_\gamma^2 & 0 & 0 \\ 0 & 0 & \sigma^2 + \sigma_\gamma^2 & \sigma_\gamma^2 \\ 0 & 0 & \sigma_\gamma^2 & \sigma^2 + \sigma_\gamma^2 \end{pmatrix}\right)$$

general

$$\mathbf{Y}_i = \mathbf{X}_i \boldsymbol{\beta} + \mathbf{Z}_i \boldsymbol{\gamma}_i + \boldsymbol{\epsilon}_i \quad (3.9)$$

the linear mixed model with fixed effects $\boldsymbol{\beta}$ and random effects $\boldsymbol{\gamma}_i$.

Assumption:

$$\left. \begin{aligned} \boldsymbol{\gamma}_i &\sim N(\mathbf{0}, \mathbf{G}), \boldsymbol{\epsilon}_i \sim N(\mathbf{0}, \boldsymbol{\Sigma}_i) \\ \boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_g, \boldsymbol{\epsilon}_1, \dots, \boldsymbol{\epsilon}_g &\text{ independent.} \end{aligned} \right\} \Rightarrow$$

$$\mathbf{Y}_i \sim N(\mathbf{X}_i \boldsymbol{\beta} + \mathbf{Z}_i \mathbf{G} \mathbf{Z}_i^\top + \boldsymbol{\Sigma}_i) \quad (\text{marginal model})$$

you don't care for individual patients, you just want to account for additional variance

$$(3.10)$$

$$\Rightarrow \mathbf{Y}_i | \boldsymbol{\gamma}_i \sim N(\mathbf{X}_i \boldsymbol{\beta} + \mathbf{Z}_i \boldsymbol{\gamma}_i, \boldsymbol{\Sigma}_i) \quad (\text{conditional model})$$

$$(3.11)$$

$$\begin{pmatrix} \boldsymbol{\gamma} \\ \boldsymbol{\epsilon} \end{pmatrix} \sim N\left(\begin{pmatrix} \mathbf{0} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \mathbf{G} & \mathbf{0} \\ \mathbf{0} & \mathbf{R} \end{pmatrix}\right)$$

$\mathbf{X}$ and $\mathbf{Z}$: known, fixed Design matrices
 $\boldsymbol{\beta}$: Vector of fixed effects
 $\boldsymbol{\gamma}$: Vector of random effects
 $\mathbf{R}$: Covariance matrix of error terms
 $\mathbf{G}$: Covariance matrix of random effects

$$\Rightarrow \mathbf{R} = \text{diag}(\boldsymbol{\Sigma}_i)$$

Estimation of LMM

Let $\boldsymbol{\vartheta}$ be a vector of all parameters, that occur in $\mathbf{G}$ and $\mathbf{R}$. $\Rightarrow$ variance of $\boldsymbol{\gamma}_i$ and $\boldsymbol{\epsilon}_i$

$$\text{Likelihood: } \ell(\boldsymbol{\beta}, \boldsymbol{\vartheta}) = -\frac{1}{2} (\ln |\mathbf{V}(\boldsymbol{\vartheta})| + (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top \mathbf{V}^{-1}(\boldsymbol{\vartheta}) (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}))$$

$$\text{where } \mathbf{V} = \mathbf{ZG}(\boldsymbol{\vartheta})\mathbf{Z}^\top + \mathbf{R}(\boldsymbol{\vartheta}).$$

$$\text{if } \boldsymbol{\vartheta} \text{ is known: } \hat{\boldsymbol{\beta}}(\boldsymbol{\vartheta}) = (\mathbf{X}^\top \mathbf{V}(\boldsymbol{\vartheta})^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1}(\boldsymbol{\vartheta}) \mathbf{Y}.$$

$$\hookrightarrow \text{Plugging in gives us the profile log-likelihood: } \ell(\boldsymbol{\vartheta}) = -\frac{1}{2} (\ln |\mathbf{V}(\boldsymbol{\vartheta})| + (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}(\boldsymbol{\vartheta}))^\top \mathbf{V}^{-1}(\boldsymbol{\vartheta}) (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}(\boldsymbol{\vartheta})))$$

$\Rightarrow$ Variance estimation biased, so we use **Restricted ML**

$$\text{REML: } \ell_R(\boldsymbol{\vartheta}) = \ell(\boldsymbol{\vartheta}) - \frac{1}{2} \ln |\mathbf{X}^\top \mathbf{V}(\boldsymbol{\vartheta})^{-1} \mathbf{X}|$$

$\hookrightarrow$ Can't compare REML, because design matrix is always different $\Rightarrow$ **Shrinkage**

Under the marginal model and conditional on $\boldsymbol{\vartheta}$, $\hat{\boldsymbol{\beta}}(\boldsymbol{\vartheta})$ follows a multivariate normal distribution with expectation value $\boldsymbol{\beta}$ and covariance matrix

$$\text{Var}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \quad (3.25)$$

Since $\mathbf{V}$ is unknown, it is replaced by the (RE)ML estimator $\mathbf{V}(\hat{\boldsymbol{\vartheta}})$.

$\mathbf{Y}_i = \mathbf{X}_i \boldsymbol{\beta} + \mathbf{Z}_i \boldsymbol{\gamma}_i + \boldsymbol{\epsilon}_i$
needs to be estimated
is random, so we can't estimate it but the variance can be estimated
random, so only variance and covariance can be estimated

GLMM

- The conditional distribution of $Y_{ij}|\gamma_i$ is from the exponential family (for example Bernoulli/Poisson)
- Conditional expected value $\mu_{ij} = \mathbb{E}(Y_{ij}|\gamma) = h(\mathbf{x}_{ij}^\top \beta + \mathbf{z}_{ij}^\top \gamma_i)$
- The random effects γ_i are normally distributed,

$$\gamma_i \stackrel{iid}{\sim} N(\mathbf{0}, \mathbf{D}).$$

GLMM with $g^{-1}(x) \neq x$

$\Rightarrow$ marginal $\mathbb{E}_\gamma(\mathbb{E}(Y_i|\eta_i)) \neq g^{-1}(\mathbf{x}_i^\top \beta)$

parameters β only have a conditional interpretation.

Assumptions:

- Distributional assumption I: Given the random effects γ , the Y_i are conditionally independent and the conditional density $f(Y_i|\gamma)$ belongs to the exponential family.
- Structural assumption: The **conditional** expected value $\mu_i = \mathbb{E}(Y_i|\gamma_i)$ is linked to the linear predictor

$$\eta_i = \mathbf{x}_i^\top \beta + \mathbf{z}_i^\top \gamma$$

through $\mu_i = g^{-1}(\eta_i)$ with known Link function $g(\cdot)$.

- Distributional assumption II: The random effects γ are normally distributed,

$$\gamma \sim N(\mathbf{0}, \mathbf{G})$$

$\Rightarrow$ no marginal perspective

Conditional: Odds-Ratio

$$\frac{\frac{\mathbb{P}(Y_{it}=1|x_{it}+1, \gamma_i)}{\mathbb{P}(Y_{it}=0|x_{it}+1, \gamma_i)}}{\frac{\mathbb{P}(Y_{it}=1|x_{it}, \gamma_i)}{\mathbb{P}(Y_{it}=0|x_{it}, \gamma_i)}} = \frac{\exp((x_{it}+1)\beta + \gamma_i)}{\exp(x_{it}\beta + \gamma_i)} = \exp(\beta)$$

Marginal: Odds-Ratio

$\Leftrightarrow$

$$\frac{\frac{\mathbb{P}(Y_{it}=1|x_{it}+1)}{\mathbb{P}(Y_{it}=0|x_{it}+1)}}{\frac{\mathbb{P}(Y_{it}=1|x_{it})}{\mathbb{P}(Y_{it}=0|x_{it})}} = \frac{\frac{\mathbb{E}_{\gamma_i}[\mathbb{P}(Y_{it}=1|x_{it}+1, \gamma_i)]}{\mathbb{E}_{\gamma_i}[\mathbb{P}(Y_{it}=0|x_{it}+1, \gamma_i)]}}{\frac{\mathbb{E}_{\gamma_i}[\mathbb{P}(Y_{it}=1|x_{it}, \gamma_i)]}{\mathbb{E}_{\gamma_i}[\mathbb{P}(Y_{it}=0|x_{it}, \gamma_i)]}} \neq \exp(\beta)$$

Interesting parameters: primarily β, ϑ , possibly ϕ
 $\Rightarrow$ Problem: find $\arg\max L(\beta, \vartheta, \phi)$, where

$$L(\beta, \vartheta, \phi) = \int \prod_{i=1}^n f(Y_i|\beta, \phi, \mathbf{b}, \vartheta) f(\gamma|\vartheta) d\gamma$$

β : coefficients for fixed effects
 γ : coefficients for random effects (Rand Var)
 ϕ : dispersion parameter
 $\mathbf{b}$: vector of coefficients for random effects
 ϑ : vector of variance params for γ

GEE Models

- allows for different dependency structures
- analogous to quasi-likelihood $\Rightarrow$ only specify expectation and variance, not distr.
- $\Rightarrow$ Quasi-score-function: $s_Q(\beta) = \mathbf{X}^\top \mathbf{D}(\beta) \Sigma^{-1}(\beta, \alpha)(\mathbf{y} - \mu)$

Additive Models

Piecewise Polynomials: $\min(x) = k_1 < \dots < k_m = \max(x)$

$\Rightarrow$ „Polynomial Splines“

Estimate separate polynomials

for $[k_j, k_{j+1}]$

$$f_j(z) = \gamma_{0j} + \gamma_{1j}z + \dots + \gamma_{lj}z^l, \quad j = 1, \dots, m-1,$$

Advantages:

- Local smoothness

- High flexibility

Disadvantages:

- Many parameters $((l+1) \cdot (m-1))$.

- Jumps

$\Rightarrow f$ then is $(l-1)$ -times continuously diff. ble

degree + intercept!

number of intervals

Truncated Power Series Basis:

$$f(z_i) = \underbrace{\gamma_1 + \gamma_2 z_i^1 + \gamma_3 z_i^2 + \dots + \gamma_{l+1} z_i^l}_{\text{global polynomial}} + \underbrace{\gamma_{l+2} (z_i - \kappa_2)_+^l + \dots + \gamma_{m+l-1} (z_i - \kappa_{m-1})_+^l}_{\text{truncated polynomials}}$$

every one of these is just an added feature basically and then the parameter has to be estimated
 $\hookrightarrow$ try to correct the global polynomial in each interval

with:

$$(z - \kappa)_+^l = \begin{cases} (z - \kappa)^l, & z \geq \kappa, \\ 0, & \text{otherwise} \end{cases}$$

$\rightarrow$ not ideal that it's $\neq 0$ on the right of the interval

$\Rightarrow m + l - 1$ coefficients in $m - 1$ intervals.

$\Rightarrow$ results in Polynomial Spline

Look at it as Linear Regression:

$$y = Z\gamma + \epsilon$$

with

$$Z = \begin{pmatrix} B_1(z_1) & \dots & B_d(z_1) \\ \vdots & & \vdots \\ B_1(z_n) & \dots & B_d(z_n) \end{pmatrix} = \begin{pmatrix} 1 & z_1 & \dots & z_1^l & \dots & (z_1 - \kappa_{m-1})_+^l \\ \vdots & \vdots & & \vdots & & \vdots \\ 1 & z_n & & z_n^l & & (z_n - \kappa_{m-1})_+^l \end{pmatrix}$$

intercept z polynomials

all of the interval functions are just new features

$$\Rightarrow \text{OLS: } \hat{\gamma} = (Z^T Z)^{-1} Z^T y$$

Regularization:

$$PLS(\lambda) := \sum_{i=1}^n \left(y_i - \sum_{j=1}^d \gamma_j B_j(z_i) \right)^2 + \lambda \sum_{j=l+2}^d (\gamma_j)^2 = \lambda b^T b$$

$\hookrightarrow$ only penalize coefficients of truncated polynomial, not global polynomial

$$\Rightarrow b \sim N(0, \frac{\sigma^2}{\lambda} \cdot I_{m-2}) \quad \text{and} \quad \text{var}(\gamma_j) = \tau$$

$$\Rightarrow \tau^2 = \frac{\sigma^2}{\lambda} \quad (\text{regularization constant from data})$$

$$\hookrightarrow \hat{\lambda} = \frac{\hat{\sigma}^2}{\hat{\tau}^2} \quad \left(\text{with } \tau^2 = 0 \Rightarrow \lambda = \infty \Rightarrow b = 0 \Rightarrow \text{no random effects} \right)$$

B-Splines

Recursive Definition of B-Splines:

$$B_j^0(z) = \mathbb{1}[\kappa_j, \kappa_{j+1}](z), \quad j = 1, \dots, d-1 = m-1$$

$$= \begin{cases} 1, & \kappa_j \leq z < \kappa_{j+1}, \\ 0, & \text{otherwise.} \end{cases}$$

In each step, 2 external knots $\kappa_{-l+1}, \kappa_{m+l}$ are added:

$$B_j^l(z) = \frac{z - \kappa_{j-l}}{\kappa_j - \kappa_{j-l}} \cdot B_{j-1}^{l-1}(z) + \frac{\kappa_{j+1} - z}{\kappa_{j+1} - \kappa_{j+1-l}} \cdot B_j^{l-1}(z)$$

example for $l=1$:
 here we go up!
 here we go down!
 overall: $\wedge$

Recursive definition of B-splines of order l

- number and position of k ; very important

Penalization: - penalize curvature: $\text{pen}(f) = \int (f''(z))^2 dz$.

For B-Splines with degree ≥ 2 and equidistant knots:

$$\text{pen}(\gamma) = \sum_{j=3}^d (\gamma_j - 2\gamma_{j-1} + \gamma_{j-2})^2 \propto \int (f''(z))^2 dz = \text{pen}(f),$$

$$= \gamma^T D^T D \gamma \quad \text{with} \quad D = \begin{pmatrix} 1 & -2 & 1 & \dots & 0 \\ 0 & 1 & -2 & 1 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 1 & -2 & 1 & \dots \end{pmatrix}$$

Properties:

- Dimension $d = m + l - 1$

- B-splines are a local basis, since each basis function is nonzero only in the area of $l+2$ knots.

- For $z \in [a, b]$, it holds that:

$$\sum_{j=1}^d B_j(z) = 1.$$

at every point the unweighted basis functions sum to 1
 (at every point there are multiple functions overlapping)

- Each basis function is bounded

- The derivative of the function $f(z) = \sum_{j=1}^d \gamma_j B_j(z)$ is given by

$$\frac{\partial f(z)}{\partial z} = l \cdot \sum_j \frac{\gamma_j - \gamma_{j-1}}{\kappa_j - \kappa_{j-1}} \cdot B_j^{l-1}(z).$$

- Matrix representation:

$$y = Z\gamma + \epsilon,$$

with

$$Z = \begin{pmatrix} B_1^l(z_1) & \dots & B_d^l(z_1) \\ \vdots & & \vdots \\ B_1^l(z_n) & \dots & B_d^l(z_n) \end{pmatrix}$$

- OLS estimation as in the linear model:

$$\hat{\gamma} = (Z^T Z)^{-1} Z^T y.$$

Additive Models 2

Smoothing Functions

- find f which minimizes $PLS(\lambda) = (y - f(z))^T (y - f(z)) + \lambda \int (f''(z))^2 dz$

- Solution through cubic splines with knots at every data point and boundary condition $f''(\kappa_1) = f''(\kappa_m) = 0$.

- General linear smoothing:

$$\hat{f}(z) = \sum_{i=1}^n s(z, z_i) y_i,$$

$$S(z) = (s(z, z_1), \dots, s(z, z_n))$$

For polynomial Splines:

$$s = Z(Z^T Z)^{-1} Z^T$$

with weights $s(z, z_i)$, i.e., $\hat{f}(z)$ is a weighted average of the observed y .
This is referred to as linear smoothing procedures.

$$\Rightarrow \text{Var}(\hat{f}(z)) = s(z)^T \underbrace{\text{Cov}(y)}_{=\sigma^2 I} s(z) = \sigma^2 s(z)^T s(z)$$

$$\Rightarrow \text{Assumption: } \hat{f}(z) \sim N(f(z), \sigma^2 s(z)^T s(z))$$

$$\Rightarrow \text{pointwise Conf-Int: } [L(z), U(z)] = \hat{f}(z) \pm z_{1-\frac{\alpha}{2}} \sigma \sqrt{s(z)^T s(z)}$$

- degrees of freedom: $df(s) = \text{trace}(s)$

- for penalized Splines: $df(S) = \text{trace}(Z(Z^T Z + \lambda K)^{-1} Z^T)$,

AIC for Smoothers

$$AIC = n \log(\hat{\sigma}^2) + 2(df + 1)$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(z_i))^2$$

minimize with respect to S

Cross validation

$$CV := \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}^{(-i)}(z_i))^2 \Rightarrow \text{Leave one out CV}$$

$\Rightarrow$ Additive models: define a function for every covariate

$$y_i = f_1(z_{i1}) + \dots + f_q(z_{iq}) + \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \epsilon_i,$$

- all functions centered around 0: $\sum_{i=1}^n f_1(z_{i1}) = \dots = \sum_{i=1}^n f_q(z_{iq}) = 0$.
- individual smoothing/penalization parameters for each covariate

Generalized Additive Models

$$\mathbb{E}(Y_i | z_i, x_i) = h(f_1(z_{i1}) + \dots + f_q(z_{iq}) + \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}) ; \quad Y_i | z_i, x_i \sim EF(\theta(\eta_i), \phi) \quad \text{with} \quad \eta_i = x_i^T \beta + \sum_j f_j(z_{ij})$$

- negative penalized log Likelihood for optimization: $l_{\text{pen}}(\beta, \gamma) = l(\beta, \gamma) - \frac{1}{2} \sum_{j=1}^q \lambda_j \gamma_j^T K_j \gamma_j$,

The following components can be included in the model:

- 1 linear effects of metric covariates x : βx
- 2 Effects of metric covariates of the form $\beta g(x)$ with known function g .
- 3 Smooth effects of the form $f(x)$ with unknown function f in basis representation
- 4 Effects of categorical covariates C with corresponding encoding: $\sum \beta_k Z_k(C)$ with accordingly defined variables $Z_k(C)$
- 5 Interactions of all combinations of the effects under 1-4.
- 6 Random effects: $z\gamma$

- Interactions: linear $\times$ linear : $x_1 \cdot x_2$
 Smooth $\times$ linear : $f(x_1) \cdot x_2$
 Smooth $\times$ categorical : $C \cdot f(x_2)$
 Smooth $\times$ smooth : $f(x_1, x_2)$

$$\mathbb{E}(Y) = x^T \beta + \sum f_j(x_j) + \text{Interaction terms} + z^T \gamma + \epsilon \quad (4.12)$$

Model Selection

- Data-Rich Situation: Training, Validation and Test data
- Data-Poor Situation: a) Approximate validation analytically (e.g. AIC/BIC)
b) Validation by CV or Bootstrap
c) do the Test theoretically

Error Decomposition

$$\begin{aligned}\text{Err}(x_0) &= \mathbb{E} \left[\left(Y - \hat{f}(x_0) \right)^2 \mid X = x_0 \right] \\ &= \sigma_\varepsilon^2 + \left[\mathbb{E} \hat{f}(x_0) - f(x_0) \right]^2 + \mathbb{E} \left[\hat{f}(x_0) - \mathbb{E} \hat{f}(x_0) \right]^2 \\ &= \sigma_\varepsilon^2 + \text{Bias}^2 \left(\hat{f}(x_0) \right) + \text{Var} \left(\hat{f}(x_0) \right) \\ &= \text{Irreducible Error} + \text{Bias}^2 + \text{Variance}.\end{aligned}$$

- Irreducible error: The variance of the target Y around its true mean $f(x_0)$, which cannot be avoided.
- Squared bias: The amount by which the mean of the estimate differs from the true mean; this is possibly due to a misspecified model.
- Variance: Deviation of $\hat{f}(x_0)$ around its mean.

Information Criteria: $-2\ell(\hat{\theta}) + \text{pen}(|\hat{\theta}|_0) \Rightarrow \text{AIC: } \text{pen}(p) = 2p ; \text{BIC: } \text{pen}(p) = \ln(n) \cdot p$

AIC as estimate of D_{KL} : $KL(g, f(\cdot, \theta)) = \mathbb{E}_g \left(\log \frac{g(\mathbf{y})}{f(\mathbf{y}|\theta)} \right) = \int \log \left(\frac{g(\mathbf{y})}{f(\mathbf{y}|\theta)} \right) \cdot g(\mathbf{y}) d\mathbf{y}.$

$$\begin{aligned}AI(f) &= \mathbb{E}_{\tilde{Y}_1, \dots, \tilde{Y}_n} \left\{ KL(g(\cdot), f(\cdot; \hat{\theta}(\tilde{Y}_1, \dots, \tilde{Y}_n))) \right\} \quad (\tilde{Y} \text{ being new data}) \\ &= \mathbb{E}_{\tilde{Y}_1, \dots, \tilde{Y}_n} \left\{ \mathbb{E}_{Y_1, \dots, Y_n} \log f(\mathbf{Y}; \hat{\theta}(\tilde{Y}_1, \dots, \tilde{Y}_n)) \right\}\end{aligned}$$

$\Rightarrow$ When estimating expectations with average Likelihood

$$-\frac{1}{n} \sum_{i=1}^n \log f(y_i; \hat{\theta})$$

we get AIC

$$\text{AIC} = -2 \sum_{i=1}^n \log f(y_i; \hat{\theta}) + 2p.$$

- AIC doesn't measure how close to the true model one can get.
- AIC measures how close one is on average, after estimating the unknown parameters.
- AIC takes variation in parameter estimation into account.

(Step AIC is greedy algorithm)

Bias Corrected AIC

$$\text{AIC}_C = -2\ell(\hat{\theta}) + 2p \left(\frac{n}{n-p-1} \right). \quad \text{recommended to use if } \frac{n}{p} < 40$$

BIC

$$\text{BIC} = -2\ell(\hat{\theta}) + \log(n)p.$$

- first Selection by AIC and then looking at p-values is overly optimistic

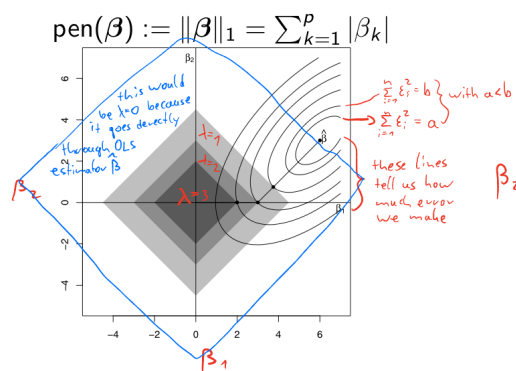
Regularization

- because not possible to search through all models with AIC

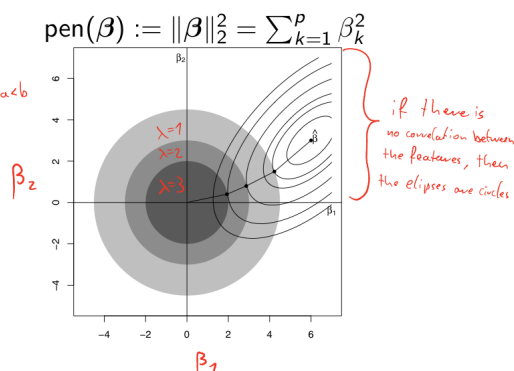
$$\Rightarrow \hat{\beta}_{PLS} = \underset{\beta}{\operatorname{argmin}} (y - X^T \beta)^T (y - X^T \beta) + \lambda \cdot \operatorname{pen}(\beta)$$

- first standardize params because not scale invariant

Lasso penalty:



Ridge penalty:



$\Rightarrow \hat{\beta}$ is OLS estimator

$$\Rightarrow \hat{\beta}_{\text{Ridge}} = (X^T X + \lambda I_p)^{-1} X^T y ; E(\hat{\beta}_{\text{Ridge}}) \neq \beta$$

$\hookrightarrow$ if X orthonormal: $\hat{\beta}_{\text{Ridge},k} = \frac{1}{1+\lambda} \hat{\beta}_{\text{OLS},k}$

$\hookrightarrow$ Looking at $MSE = \text{Bias}^2 + \text{Variance}$: $\lambda=0 \Rightarrow \text{Bias}=0$; $\lambda \rightarrow \infty \Rightarrow \text{Variance}=0$

$$\Rightarrow \hat{\beta}_{\text{Lasso}} := \underset{\beta}{\operatorname{argmin}} \frac{1}{2} (y - X^T \beta)^T (y - X^T \beta) + \lambda \sum_{k=1}^p |\beta_k| \quad (\text{can only be solved numerically})$$

$\hookrightarrow$ if X orthonormal: $\hat{\beta}_{\text{Lasso},k} = \operatorname{sign}(\hat{\beta}_{\text{OLS},k}) \max(0, |\hat{\beta}_{\text{OLS},k}| - \lambda)$

SCAD & MCP to not introduce such a big bias (like L1 for small β and L2 for large β)

$\Rightarrow$ to find best λ_0 , use Cross Validation (with standard dev $\hat{\sigma}$)

$\hookrightarrow$ often still too complex, therefore find largest λ with $MSE(\lambda) < MSE(\lambda_0) + \hat{\sigma}$

Group Lasso

- for example remove all categories of one variable

- for groups a, b, c, d : $\beta_g = \begin{pmatrix} \beta_b \\ \beta_c \\ \beta_d \end{pmatrix}$: $\operatorname{pen}(\beta) = \sum_{g=1}^G \|\beta_g\|_2 = \sum_{g=1}^G \sqrt{\beta_g^T \beta_g}$

Fused Lasso

$$\operatorname{pen}(\beta) = \sum_{k=2}^K |\beta_k - \beta_{k-1}| \Rightarrow \text{shrink difference between coefficients}$$

Lasso for GLM

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} \left(\sum_{i=1}^n -\ell(\beta; y_i, x_i) + \lambda \sum_{j=1}^p |\beta_j| \right)$$

Distributional Regression

- with most Regression models we assume homoskedasticity, even with non-linear models

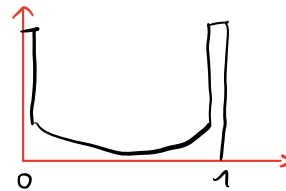
- Heteroskedastic Model: $Y|x \sim N(\mu(x), \sigma(x)^2)$
 $\mu(x) = \beta_{0\mu} + f_\mu(x)$
 $\sigma(x) = \exp(\beta_{0\sigma} + f_\sigma(x))$
 Link-Function that ensures we get a positive value

- Basic Idea:
- Every distribution parameter can be modeled, not just the mean ("location")
 - Covariates define the entire shape of the conditional distribution.

GAMLSS - GAM for Location, Scale, Shape

$$Y|x \sim D(\mu(x), \sigma(x), \nu(x), \tau(x)) \text{ with } \left. \begin{array}{l} g_\mu(\mu) = \eta_\mu = \beta_{0\mu} + \sum_{j=1}^{p_\mu} f_{j\mu}(x_j) \quad \text{"location"} \\ g_\sigma(\sigma) = \eta_\sigma = \beta_{0\sigma} + \sum_{j=1}^{p_\sigma} f_{j\sigma}(x_j) \quad \text{"scale"} \end{array} \right| \begin{array}{l} g_\nu(\nu) = \eta_\nu = \beta_{0\nu} + \sum_{j=1}^{p_\nu} f_{j\nu}(x_j) \quad \text{"skewness"} \\ g_\tau(\tau) = \eta_\tau = \beta_{0\tau} + \sum_{j=1}^{p_\tau} f_{j\tau}(x_j) \quad \text{"kurtosis"} \end{array}$$

↳ also implements 0-1-inflated models, etc. :



$$f(y) = \pi_1(x) \cdot \mathbb{1}(y=0) + \pi_2(x) \cdot f_{\text{beta}}(y) + \pi_3(x) \cdot \mathbb{1}(y=1)$$

Quantile Regression

- Idea of quantile regression: directly model (conditional) quantiles, without going through a parametric distribution.

↳ with other models, we first estimate distribution and then using that the quantiles

- Let $Q_\tau(y)$ be the τ -quantile of the distribution of y . $\Rightarrow \mathbb{P}(Y < Q_\tau(y)) = F_Y(Q_\tau(y)) = \tau$
- Then we model:

$$Q_\tau(y|x) = \beta_{\tau 0} + \beta_{\tau 1}x_1 + \dots + \beta_{\tau p}x_p = \mathbf{x}^\top \beta_\tau$$

or also

$$Q_\tau(y|x) = \mathbf{x}^\top \beta_\tau + f_{\tau(1)}(z_1) + \dots + f_{\tau(j)}(z_j)$$

$\Rightarrow$ Estimate τ -Quantile with: $q_\tau = \arg \min \mathbb{E}_Y(w_\tau(y, q)|y - q|) = \arg \min_q \left(\int (w_\tau(y, q)|y - q|f(y)dy) \right)$

$$\hookrightarrow w_\tau(y, q) = \begin{cases} 1 - \tau & \text{for } y < q \\ \tau & \text{for } y > q \\ 0 & \text{for } y = q \end{cases}$$

$\Rightarrow$ as optimization problem for observed data:

$$\sum_{i=1}^n \rho_\tau(y_i - Q_\tau(y|x)) \rightarrow \min \quad \text{with} \quad \rho_\tau(u) = \begin{cases} u \cdot \tau & \text{for } u \geq 0 \\ u \cdot (\tau - 1) & \text{for } u < 0 \end{cases} \quad \} \text{Pinball loss}$$

Estimation: $Q_\tau(y_i|x_i) = \mathbf{x}_i^\top \beta_\tau$: $\hat{\beta} = \arg \min_{\beta} \sum_{i=1}^n \rho_\tau(y_i - \mathbf{x}_i^\top \beta)$

↳ Quantiles are estimated individually $\Rightarrow$ Quantile crossing !!!

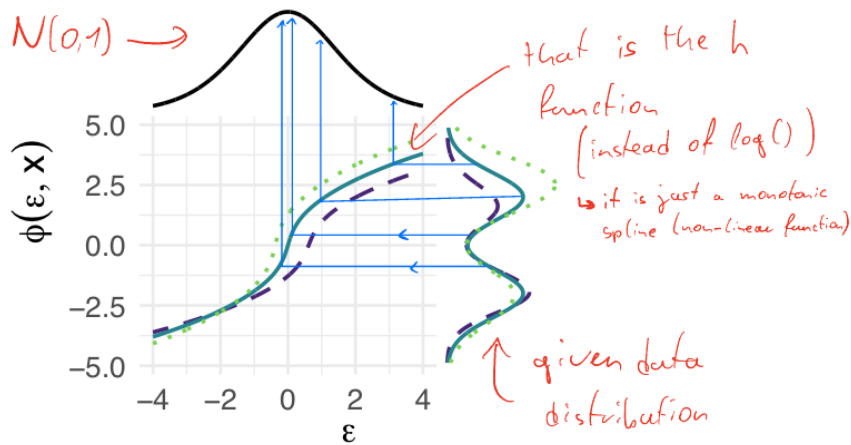
Transformation Models

- models the function for y , so that after applying that function, we know the distribution of y

↳ simple case: linear model in $\log(y)$

⇒ function needs to be monotone

$$\hat{\beta} = \arg \min_{\beta} \sum_{i=1}^n \rho_{\tau}(y_i - \mathbf{x}_i^{\top} \beta)$$



- CDF can be modelled directly: $F_Y(y) := \mathbb{P}_Y(Y \leq y) = \mathbb{P}_{h(Y)}(h(Y) \leq h(y)) =: \mathbb{P}_Z(Z \leq z) =: F_Z(z)$,

- for PDF we need Density transformation: $f_Y(y) = f_Z(h(y)) \cdot \left| \frac{\partial h(y)}{\partial y} \right|$

- Log-Likelihood for iid case: $\log f_Y(\mathbf{y}) = \log \prod_{i=1}^n f_Y(y_i) = \log \prod_{i=1}^n \left\{ f_Z(h(y_i)) \cdot \left| \frac{\partial h(y_i)}{\partial y_i} \right| \right\}$

Functional data

Properties of functional data:

- The data varies over its range of values (hereafter: time).
- Underlying data-generating process assumed to be smooth
- Curve is observed at discrete time points in time domain $\mathcal{T}$
- One observation $\hat{=}$ One curve (= several single measurements)
- n observations = n observed curves

$\Rightarrow$ observed as $w_i(t) = x_i(t) + \varepsilon_i(t)$
with noise $\varepsilon_i(t)$

- Consider functional random variable $X(t)$, with $t \in \mathcal{T}$ and $\mathcal{T} \subseteq \mathbb{R}$.
- We observe samples $x_i(t)$, $i = 1, \dots, n$

- Functional mean: $\hat{\mu}_X(t) = \bar{x}(t) = n^{-1} \sum_{i=1}^n x_i(t)$
 - Functional variance: $\hat{\sigma}_X^2(t) = (n-1)^{-1} \sum_{i=1}^n [x_i(t) - \bar{x}(t)]^2$
 - Functional covariance: $\hat{\sigma}_X(t_1, t_2) = (n-1)^{-1} \sum_{i=1}^n [x_i(t_1) - \bar{x}(t_1)][x_i(t_2) - \bar{x}(t_2)]$
- } Operations per time point t

2 ways of centering

- 1) subtract mean per timepoint from the curves at that time point:

$$x_i^*(t) = x_i(t) - \bar{x}(t)$$

- 2) subtract the mean per observation from each measurement of that observation

$$\tilde{x} = x_i - \int \frac{x_i(t)}{t} dt$$

Functional PCA

- find dimensions that explain most of the variance in the data $\Rightarrow$ functional eigenvectors

The zero-mean square-integrable stochastic process $X(t) - \mu(t)$ can be approximated as follows:

$$X(t) - \mu(t) = \sum_{k=1}^{\infty} \xi_k \varphi_k(t), \quad (9.1)$$

$\downarrow$ centered
 $\hookrightarrow$ these are the eigen functions

$\Rightarrow$ with $\|\varphi\| = 1$

$\Rightarrow \varphi$ orthogonal:

$$\langle \varphi_1, \varphi_2 \rangle = \int \varphi_1(t) \varphi_2(t) dt = 0$$

where ξ_k are the FPC scores and $\varphi_k(t)$ are the FPCs.

$\Rightarrow \varphi_1$ explains the most variance possible and then φ_2 most of remaining variance

- FPCA can be used for modeling, for example:
 - If **covariates are functions, reduce to scores** via FPCA and treat scores as new features in, e.g., a GAM
 - If both covariates and response are functions and measured on the same domain, we could regress scores on scores
 - Functional Regression models

Functional Regression models

1 Scalar-on-Function Regression (SOFR)

2 ^{Target} Function-on-Scalar Regression (FOSR)

3 ^{Input} Function-on-Function Regression (FOFR)

1) Scalar-on-Function

Options for modeling functional covariates:

- Reduce via FPCA, then use a "scalar model"
- Incorporate functional covariate in the regression model as-is:
Analogous to scalar regression, now with infinitely many covariate values $x(t), t \in \mathcal{T}$
 - Linear functional effects: $x(t)\beta(t)$
 - Non-linear functional effects: $f(x(t), t)$

Example: Scalar-on-function linear model:

$$\mathbb{E}(Y|\mathbf{x}) = \beta_0 + x_1\beta_1 + \underbrace{\int_{\mathcal{T}} x_2(t)\beta_2(t)dt}_{\text{scalar}}$$

$\Rightarrow$ Every time point of x_2 has a linear effect and we finally integrate over all t

But we can't calculate integral because we don't observe all time points $\Rightarrow$ that's why we approximate it

$$\int_{\mathcal{T}} x_2(t)\beta_2(t)dt \approx \sum_{r=1}^R \Delta_r x_2(t_r)\beta_2(t_r)$$

2) Function-on-Scalar $Y(t) = x\beta(t) + \epsilon(t), \epsilon(t) \sim N(0,1)$

- Functional Intercept model: $g(\mathbb{E}(Y(t))) = \beta_0(t) \Rightarrow$ at every t mean of all data curves

- with categorical x : $g(\mathbb{E}(Y(t)|x)) = \beta_0(t) + \beta_1(t)I_{\text{medium}}(x) + \beta_2(t)I_{\text{large}}(x) \Rightarrow$ "small" in intercept

$\Rightarrow$ Estimation: $\Rightarrow$ Minimizing the residuals over all time points, i.e.,

$$\operatorname{argmin}_{\beta(t)} \int_{\mathcal{T}} (Y(t) - x\beta(t))^2 dt = \operatorname{argmin}_{\beta(t)} \int_{\mathcal{T}} (\epsilon(t))^2 dt \approx \operatorname{argmin}_{\beta(t)} \sum_{r=1}^R \Delta_r \epsilon(t_r)^2.$$

$\hookrightarrow$ in practice because finite data points

3) Function-on-Function

- Response and covariates do not need to be measured on the same function domain, e.g., $Y(t), t \in \mathcal{T}$ and $x(s), s \in \mathcal{S}$

- Consequently, we need to define the functional relationship between the model in- and outputs:

- Concurrent effect: *only current state of features have impact on outcome*
 $g(\mathbb{E}(Y(t)|x)) = \beta_0(t) + x(t)\beta(t)$

- Linear effect: *whole feature observation (past & future) have impact on current outcome*
 $g(\mathbb{E}(Y(t)|x)) = \beta_0(t) + \int_{\mathcal{S}} x(s)\beta(s, t)ds$

- Historical linear effect:

$$g(\mathbb{E}(Y(t)|x)) = \beta_0(t) + \int_{s < t - \text{lag}} x(s)\beta(s, t)ds \quad \leftarrow \text{only past values can}$$

- General linear effect with flexible integration limits: *impact my current measurement*

$$g(\mathbb{E}(Y(t)|x)) = \beta_0(t) + \int_{u(t)}^{o(t)} x(s)\beta(s, t)ds$$

- Non-linear effect:

$$g(\mathbb{E}(Y(t)|x)) = \beta_0(t) + \int f(x(s), s, t)ds$$

Random intercept model is centered around 0 and shows the shift as estimate

$\Rightarrow R$ function plfr

$$c() = \frac{S^2 C()}{S u S v}$$

Copula

- Model for dependence, can be combined with any marginal distr.

Sklar's Theorem

- $X = (X_1, \dots, X_d)$ with joint distr. F and marginals $F_1, \dots, F_d$

$$\Rightarrow \exists C : F(x) = C(F_1(x_1), \dots, F_d(x_d))$$

- C is joint cdf of $U = (F_1(X_1), \dots, F_d(X_d))$; $F_i(X_i) \sim U(0,1)$

$$\hookrightarrow \text{Bew: } P(U \leq u) = P(F_Y(Y) \leq u) = P(Y \leq F_Y^{-1}(u)) = F_Y(F_Y^{-1}(u)) = u$$

- Empirical(γ : copula $\approx$ distribution of (normalized) ranks. ; $\text{rank}(\gamma_i) = n \cdot \hat{F}(\gamma_i)$

$$\begin{aligned} F(x_1, \dots, x_d) &= \Pr\{X_1 \leq x_1, \dots, X_d \leq x_d\} \\ [F_j \text{ increasing}] &= \Pr\{F_1(X_1) \leq F_1(x_1), \dots, F_d(X_d) \leq F_d(x_d)\} \\ &= C\{F_1(x_1), \dots, F_d(x_d)\} \end{aligned}$$

- This argument works for **any distribution**, not just continuous ones!

Sklar for densities

Any multivariate density $f \sim (X_1, \dots, X_d)$ can be decomposed into

$$f(x) = c(F_1(x_1), \dots, F_d(x_d)) \times \prod_{j=1}^d f_j(x_j), \quad \text{for all } x \in \mathbb{R}^d.$$

$\Rightarrow c(\dots)$ adds dependence

$$f(x_1, \dots, x_d) = \frac{\partial^d}{\partial x_1 \dots \partial x_d} F(x_1, \dots, x_d)$$

$$[\text{Sklar}] = \frac{\partial^d C(F_1(x_1), \dots, F_d(x_d))}{\partial x_1 \dots \partial x_d}$$

$\Rightarrow$ Bew:

$$[\text{Chain rule}] = \frac{\partial^d C(F_1(x_1), \dots, F_d(x_d))}{\partial F_1(x_1) \dots \partial F_d(x_d)} \times \prod_{j=1}^d \frac{\partial F_j(x_j)}{\partial x_j}$$

$$= c(F_1(x_1), \dots, F_d(x_d)) \times \prod_{j=1}^d f_j(x_j).$$

Constructing a Copula with inversion of Sklar

- $C(u) = F(F_1^{-1}(u_1), \dots, F_d^{-1}(u_d))$ because $u_j = F_j(x_j)$

$\hookrightarrow$ Gaussian Copula: $C(u) = \Phi(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_d))$

- Construction by generator function $\psi: [0,1] \rightarrow [0,\infty]$, $\psi(1)=0$: $C(u) = \psi^{[-1]}(\psi(u_1) + \dots + \psi(u_d))$.

- Clayton copula: $\psi(u) = (u^{-\delta} - 1)/\delta \Rightarrow C(u_1, u_2) = (u_1^{-\delta} + u_2^{-\delta} - 1)^{-\frac{1}{\delta}}$
- Gumbel copula: $\psi(u) = (-\ln u)^\delta \Rightarrow C(u_1, u_2) = \exp[-\{(-\ln u_1)^\delta + (-\ln u_2)^\delta\}^{\frac{1}{\delta}}]$

Rotations/Reflections

- 90 degree: $c_{90}(u_1, u_2) := c(1 - u_1, u_2)$
- 180 degree: $c_{180}(u_1, u_2) := c(1 - u_1, 1 - u_2)$
- 270 degree: $c_{270}(u_1, u_2) := c(u_2, 1 - u_1)$

Estimation:

Given iid data $X_1, \dots, X_n \in \mathbb{R}^d$:

1) Estimate marginal distributions $\hat{F}_1, \dots, \hat{F}_d$

- parametric: $\hat{F}_j(x) = F_j(x; \hat{\theta}_j)$
- nonparametric: $\hat{F}_j(x) = \sum_{i=1}^n \mathbb{1}(X_{i,j} \leq x)/(n+1)$

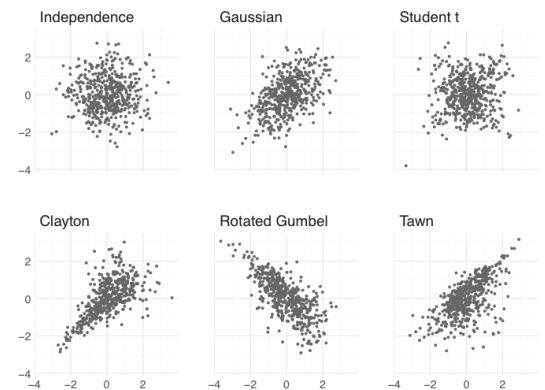
2) Compute pseudo-observations for the copula:

$$\hat{U}_i = (\hat{U}_{i,1}, \dots, \hat{U}_{i,d}) = (\hat{F}_1(X_{i,1}), \dots, \hat{F}_d(X_{i,d})).$$

3) Estimate copula parameters η by pseudo-maximum likelihood

$$\hat{\eta} = \arg \max_{\eta} \sum_{i=1}^n \log c(\hat{U}_i; \eta).$$

Gaussian margins + various copulas



Copula 2

Sampling: from model F with Copula C and marginals F_i :

- 1 Sample from the copula C : $\mathbf{U} = (U_1, \dots, U_d)$
- 2 Apply the inverse PIT: $\mathbf{X} := (F_1^{-1}(U_1) \dots F_d^{-1}(U_d))$

Dependence measures

Pearson: $\rho(X_1, X_2) = \text{Cor}(X_1, X_2) = \frac{\text{Cov}(X_1, X_2)}{\sqrt{\text{Var}(X_1)\text{Var}(X_2)}}$

- bad: only linear dependence
- depends on marginal distr.

Better: Spearman: $\rho_S(X_1, X_2) = \text{Cor}(F_1(X_1), F_2(X_2))$

- empirically $F_i(X_i)$ would be ranks

$$= 12 \int_{[0,1]^2} u_1 u_2 c(u_1, u_2) du_1 du_2 - 3 \quad (\text{expectation of } u_1, u_2)$$

Better: Kendall: $\tau(X_1, X_2) = P((X_{11} - X_{21})(X_{12} - X_{22}) > 0) - P((X_{11} - X_{21})(X_{12} - X_{22}) < 0) = 4 \int_{[0,1]^2} C(u_1, u_2) c(u_1, u_2) du_1 du_2 - 1$
where (X_{11}, X_{12}) and (X_{21}, X_{22}) are independent copies of (X_1, X_2) .

=> how many concordance (↗) and discordance: (↘)

Copula	Kendall's τ
Gaussian	$\tau = \frac{2}{\pi} \arcsin(\rho)$
Student t	$\tau = \frac{2}{\pi} \arcsin(\rho)$
Clayton	$\tau = \frac{\delta}{\delta+2}$
Gumbel	$\tau = 1 - \frac{1}{\delta}$

Important: independence $\Rightarrow \begin{cases} \tau = 0 \\ \rho_P = 0 \\ \rho_S = 0 \end{cases}$

Best: Mutual Information: $MI \equiv KL(f || f_1 \cdot f_2)$
 $= E_f \left(\log \left(\frac{f(x_1, x_2)}{f_1(x_1) f_2(x_2)} \right) \right)$
 $= E_c \left(\log(c(u_1, u_2)) \right) = H(c) \quad (\text{Entropy of Copula})$

=> $MI=0 \Leftrightarrow X_1, X_2$ independent

Tail dependence

- The upper tail dependence of the variables is defined by => looks asymmetric but order doesn't matter

$$\lambda_u = \lim_{u \rightarrow 1} P(Y_1 \geq F_1^{-1}(u) | Y_2 \geq F_2^{-1}(u)) = \lim_{u \rightarrow 1} \frac{1 - 2u + C(u, u)}{1 - u}$$

- The lower tail dependence is given by

$$\lambda_l = \lim_{u \rightarrow 0} P(Y_1 \leq F_1^{-1}(u) | Y_2 \leq F_2^{-1}(u)) = \lim_{u \rightarrow 0} \frac{P(Y_1 \leq F_1^{-1}(u), Y_2 \leq F_2^{-1}(u))}{P(Y_2 \leq F_2^{-1}(u))} = \lim_{u \rightarrow 0} \frac{C(u, u)}{u}$$

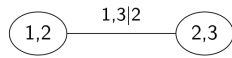
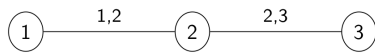
Copula	λ_u	λ_l
Gaussian	0	0
Student t	$2t_{\nu+1} \left(-\sqrt{\nu+1} \sqrt{\frac{1-\rho}{1+\rho}} \right)$	$2t_{\nu+1} \left(-\sqrt{\nu+1} \sqrt{\frac{1-\rho}{1+\rho}} \right)$
Clayton	0	$2^{-1/\delta}$
Gumbel	$2 - 2^{1/\delta}$	0

Copula 3

Pair Copula Decomposition for dimensions > 2

$$c \sim (U_1, U_2, U_3): c(u) = c_{1,2}(u_1, u_2) \cdot c_{2,3}(u_2, u_3) \cdot c_{1,3;2}(C_{1|2}(u_1|u_2), C_{3|2}(u_3|u_2); u_2)$$

Vine Copulas:



$$\Rightarrow f = f_1 \times f_2 \times f_3 \times c_{1,2} \times c_{2,3} \times c_{1,3|2}$$

$\hookrightarrow$ dimensions: # of possible vine structures: $d! \times 2^{\binom{d-2}{2}/2}$

of pairs of copulas: $\binom{d}{2}$

$\hookrightarrow$ parameter estimation sequentially

- Dissman's algorithm: at every level,
 - 1 (Structure selection) Construct maximum spanning tree based on empirical Kendall's τ of pseudo-observations.
 - 2 (Parameter estimation) Estimate pair-copula parameters for some copula families.
 - 3 (Model selection) Choose family with highest AIC/BIC.

Causality, DAGs, IV

Experimental studies: adhere to randomly assigned treatment (randomized (controlled) trials).

- ⇒ minimize challenges of causal identification arising from confounding;
- ⇒ may be susceptible to sample selection bias and other imperfections;
- ⇒ sample sizes may be limited by both cost and effort.

Observational studies: data available on more comprehensive, representative real-world conditions.

- ⇒ larger sample size;
- ⇒ confounding often poses a challenge to causal identification;
- ⇒ sample sizes may be limited by both cost and effort.

Causal Roadmap:

- 1) Well-defined and unambiguous scientific question.
- 2) Specify a target causal quantity.
- 3) Specify a causal model representing real background knowledge about data-generating mechanism.
- 4) Specify the observed data and their link to the causal model.
- 5) Assess identifiability.
- 6) Commit to a statistical model and estimand.
- 7) Estimate.
- 8) Interpret.

- interested in causal relationship between X & Y , but we have unobservable & observable confounders U & Z

$$\Rightarrow \int f_{Y|X,Z,U} f_{Z,U} dz du = \int \frac{f_{Y,X,Z,U}}{f_{X|Z,U}} dz du \neq f_{Y|X}$$

↳ only equal if X is ind. from U & Z

⇒ because then $f_{X|Z,U} = f_X$

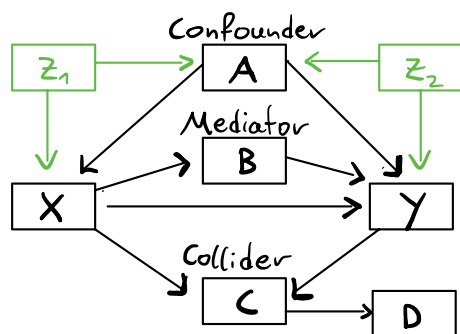
⇒ achieved with randomization

Exogeneity: - in linear model $Y = \beta_0 + \beta_1 X + \varepsilon$: $\text{Cov}(X, \varepsilon) = 0 \Rightarrow E(\varepsilon|X) = 0$

Endogeneity: - but with $Y = \beta_0 + \beta_X X + \underbrace{\beta_U U + \varepsilon}_{=\tilde{\varepsilon}}$: if $X \not\perp U \Rightarrow X \not\perp \tilde{\varepsilon} \Rightarrow \frac{\delta Y}{\delta X} = \beta_X + \frac{\delta \tilde{\varepsilon}}{\delta X}$ bias

Directed Acyclic Graphs (DAG)

- A DAG is a graph that provides a visual representation of causal relationships among a set of variables.
- Each arrow in a DAG represents a causal relationship.
- DAGs can be used to select variables to include in regression models, specifically with the aim of estimating causal relationships from non-randomized studies, which are subject to confounding.



⇒ $y = \beta_0 + \beta_1 \cdot c$ doesn't make sense because Y influences C

Confounding

Confounding is the bias that appears, if one or more variable(s) have an effect on both the outcome and the intervention of interest.

Confounding (back door path):

In a DAG, a back-door path is defined as a path from X to Y that starts with an arrow into X . This path generates confounding for the effect of X on Y .

- Conditioning on A blocks path

- Conditioning on B blocks path

- Conditioning on C or D opens path

- Conditioning on A not enough, because backdoor path $X \leftarrow Z_1 \rightarrow A \leftarrow Z_2 \rightarrow Y$ is not blocked

d-separate
= no active path

A set of variables S is said to block, or d-separate, a path p in a DAG if either:

- 1) p contains a chain $X \rightarrow Z \rightarrow Y$ or a fork $X \leftarrow Z \rightarrow Y$ such that Z is in S or
- 2) p contains an inverted fork $X \rightarrow Z \leftarrow Y$ such that neither Z nor a descendant of Z is in S

Special case:

If S blocks all paths from X to Y , then X and Y are independent conditional on S . We write $X \perp\!\!\!\perp Y \mid S$.

Causality, DAGs 2

Instrumental Variable (IV)

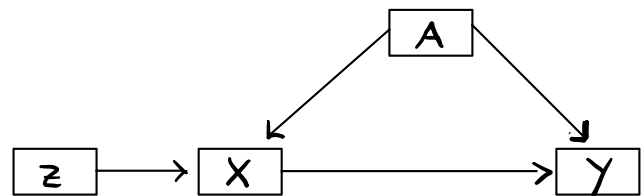
Backdoor Criterion => blocking back-door paths

A backdoor path from X to Y is defined as a path starting with an edge ending in X. A set of variables S satisfies the back-door criterion with respect to two variables X and Y in a DAG if:

- 1 S blocks all back-door paths from X to Y and
- 2 No element of S is a descendant of X.

Methods for Drawing Causal Conclusions

- Randomized experiment: X is chosen randomly for the units (e.g. treatment and control if X is binary). Many designs are possible (design of experiments).
- Natural experiment: X is chosen to be independent from other covariates. It is similar to random.
- Perform multiple regression with confounders or all necessary variables deduced from a DAG. If all confounders are observable, this is a good strategy.
- Instrumental variables (IV).
- Propensity score matching.



- i) Z is uncorrelated with the residual $\tilde{\epsilon}$. $\text{Cov}(Z, A) = 0$
 - ii) Z is correlated with X. $\text{Cov}(Z, X) > 0$
- $\tilde{\epsilon} = A \cdot \beta + \epsilon$

=> Idea: if we change Z, we change X, but not A

- we want to see how Y changes with X while U is fixed:

$$\frac{\partial Y}{\partial X} | (U = u) = \frac{\partial Y / \partial Z}{\partial X / \partial Z} | (U = u) = \frac{\partial Y / \partial Z}{\partial X / \partial Z}$$

Because Z is independent of U, we can omit the condition $U = u$.

IV-Estimator:

We fit a regression of Y on Z and X on Z, giving the two estimates $\hat{\beta}_{YZ}$ and $\hat{\beta}_{XZ}$.

$$\Rightarrow \hat{\beta}_X = \frac{\hat{\beta}_{YZ}}{\hat{\beta}_{XZ}}$$

Identical to: Two Stage Least Squares (2SLS)

1) $\hat{X} = Z(Z^T Z)^{-1} Z^T X$

2) $\hat{\beta}_X = (\hat{X}^T \hat{X})^{-1} \hat{X}^T Y$

Propensity Score matching

Relevant effect of a treatment T:

- $Y_i(0)$: Outcome of person i with treatment 0.
- $Y_i(1)$: Outcome of person i with treatment 1.

Causal treatment effect:

$$\mathbb{E}(Y_i(1) - Y_i(0)). \rightarrow \text{Average over all observations}$$

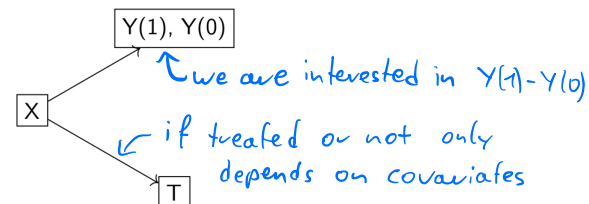
We are interested in:

Average Treatment Effect on the Treated (ATT)

$$\tau = \mathbb{E}(Y(1) | T = 1) - \mathbb{E}(Y(0) | T = 1)$$

But we can only observe: $\mathbb{E}(Y_i(1) | T_i = 1) - \mathbb{E}(Y_i(0) | T_i = 0) = \tau + \underbrace{\mathbb{E}(Y(0) | T = 1) - \mathbb{E}(Y(0) | T = 0)}_{\text{bias}}$

Idea: In the presence of covariates X, we make the assumption that the potential outcomes $Y(0)$ and $Y(1)$ are conditionally independent of the treatment variable T, given our knowledge of X.



=> 1) logistic Reg. to estimate if someone is treated based on X $\mathbb{P}(T = 1 | X) =: p(X)$,

2) match similar $p(X)$ ("digital twins")

3) Check that balance/distribution of observed covariates is similar in both groups.

4) Use methods for randomized experiments to estimate treatment effect.

Assumptions:

- **Positivity:** Each subject in the study must have had (theoretical) probability > 0 of being in the opposite group.
- **Stable-unit-treatment-value:** The effect of the treatment on the outcome does not depend on the treatment status of other subjects.
- **Exchangeability:** All relevant confounders were observed.

Survival Analysis

special about survival analysis: - covariates can change over observation time

- covariate effects can change

- subject dropout before event

↳ instead of observing T , we only know $T > c$ (right-censoring)

• Density: $f_T(t) := f(t)$

• Cumulative distribution function: $F_T(t) := F(t) := P(T \leq t)$

• Survival function: $S_T(t) := S(t) := P(T > t) = 1 - F(t)$

• Hazard rate: $h_T(t) := h(t) := \lim_{\Delta t \searrow 0} \frac{1}{\Delta t} P(t \leq T < t + \Delta t | T \geq t)$ instantaneous chance of death

• Cumulative hazard rate: $H_T(t) := H(t) := \int_0^t h(u) du$

- Relations: $h(t) = \frac{f(t)}{1 - F(t)} = \frac{f(t)}{S(t)}$

$$S(t) = \exp\left(-\int_0^t h(u) du\right)$$

$$f(t) = -\frac{d}{dt} S(t)$$

Weibull distr $T \sim WB(\alpha, \lambda)$

• Lifetime of objects with a hazard proportional to a power of time

Hazard rate:

$$h(t) = \lambda \cdot \alpha (\lambda t)^{\alpha-1}$$

Survival function:

$$S(t) = \exp(-(\lambda t)^\alpha)$$

Density:

$$f(t) = \lambda \cdot \alpha (\lambda t)^{\alpha-1} \exp(-(\lambda t)^\alpha)$$

Censoring

$T_i \sim F_T$ duration times

$C_i \sim F_C$ censoring times,

and we observe: $Y_i = \min(T_i, C_i)$, with $D_i = \begin{cases} 1, & Y_i = T_i \\ 0, & Y_i = C_i \end{cases}$

Kaplan-Meier Estimator

- discrete hazard rates for interval $[t_{(k-1)}, t_{(k)})$ for non-censored data:

$$h_k = P(T \in [t_{(k-1)}, t_{(k)}) | T \geq t_{(k-1)})$$

- Probability to survive longer than k -th interval: $P_k = P(T \geq t_{(k)})$

Since $P(T \geq t_{(k)}) = P(T \geq t_{(k)} | T \geq t_{(k-1)}) P(T \geq t_{(k-1)})$ for all k

and $P(T \geq 0) = 1$ we get: $P_k = p_k \cdot \dots \cdot p_1 = (1 - h_k) \cdot \dots \cdot (1 - h_1)$.

- Estimation of Survival Function:

$$\hat{S}(t) = \begin{cases} 1 & t < t_{(1)} \\ \prod_{t_{(k)} \leq t} (1 - d_k/n_k) & t \geq t_{(1)} \end{cases}$$

• d_k number of events at time point $t_{(k)}$

• n_k number of individuals at risk just before time point $t_{(k)}$

Likelihood Estimation

No censoring: - density $f(y|\theta)$; hazard rate $h(y|\theta)$

$$L(\theta) = \prod_{i=1}^n f(Y_i = y_i | \theta) = \prod_{i=1}^n f(y_i | \theta)$$

Censoring: • uncensored observations, i.e. $D_i = 1$ and $Y_i = T_i$ and $T_i \leq C_i$

• censored observations, i.e., $D_i = 0$ and $Y_i < T_i$ and $C_i < T_i$

and we assume random censoring $T_i \perp C_i$

$$\Rightarrow \text{Density: } f(y_i, d_i | \theta, \psi) = f_T(y_i | \theta)^{d_i} S_C(y_i | \psi)^{d_i} \cdot f_C(y_i | \psi)^{1-d_i} S_T(y_i | \theta)^{1-d_i}$$

$\underbrace{\hspace{10em}}$
P that event
occurs before censoring

$\underbrace{\hspace{10em}}$
P that event
occurs after censoring

$$\Rightarrow L(\theta) = c \prod_{i=1}^n f(y_i | \theta)^{d_i} S(y_i | \theta)^{1-d_i} = c \prod_{i=1}^n h(y_i | \theta)^{d_i} \exp\left(-\int_0^{y_i} h(s | \theta) ds\right)$$

$$\text{and } l(\theta) = \log(c) + \sum_{i=1}^n (d_i \log h(y_i | \theta) - H(y_i | \theta))$$

Survival Analysis 2

Parametric transformation models

- transformation $g(T)$

$$\hookrightarrow \text{i.e. } \log(T) = \mathbf{x}'\beta + \sigma\epsilon \iff T = \exp(\mathbf{x}'\beta) \cdot \exp(\sigma\epsilon) \\ = \exp(\beta_0) \cdot \exp(\beta_1 x_1) \dots \exp(\beta_p x_p) \exp(\sigma\epsilon)$$

$\Rightarrow$ multiplicative effect on duration time

$\Rightarrow$ when increasing x_1 by 1 we multiply T by $\exp(\beta_1)$

Distribution of ϵ or $\log(T)$	Distribution of T
$\epsilon \sim N(0, 1)$	lognormal
$\epsilon \sim$ generalized extreme value	Weibull (Exp.)
$\epsilon \sim$ logistic	log-logistic
log-Gamma	Gamma
log-gen. Gamma	generalized Gamma

ML-Estimation:

$$L(\theta) = \prod_{i=1}^n (f(y_i|\theta, \mathbf{x}_i))^{d_i} (S(y_i|\theta, \mathbf{x}_i))^{1-d_i}$$

$$l(\theta) = \sum_{i=1}^n d_i \log(f(y_i|\theta, \mathbf{x}_i)) + (1 - d_i) \log(S(y_i|\theta, \mathbf{x}_i))$$

$\Rightarrow$ Maximization gives $\hat{\beta}_{ML} \overset{a}{\sim} N(\beta, \Sigma_{\hat{\beta}})$

Hypothesis testing

- θ is vector of parameters β (and σ)

$$\Rightarrow H_0: A\theta = c, H_1: A\theta \neq c$$

- **Likelihood-Ratio Statistic:** $l_q = 2(\ln L(\hat{\theta}) - \ln L(\tilde{\theta}))$ ($\hat{\theta}$ ML-Estimator under H_1 ; $\tilde{\theta}$ under H_0)
- **Wald Statistic:** $W = (A\hat{\theta} - c)'[A \text{Cov}(\hat{\theta})A']^{-1}(A\hat{\theta} - c)$.

$\Rightarrow$ both under H_0 $\chi^2(m)$ distr ($m = \text{rg}(A)$)

$$-AIC = -2\ell(\hat{\theta}_{ML}) + 2 \cdot df$$

Semi-parametric Regression

- no or weak assumptions about distribution of data

Cox Proportional Hazards model

$$h(t, \mathbf{x}) = h_0(t) \exp(\mathbf{x}'\beta) = h_0(t) \cdot \exp(\beta_1 x_1) \cdot \dots \cdot \exp(\beta_p x_p)$$

$h_0(t)$: baseline hazard rate identical for all individuals,
a nuisance parameter!

β : parameter vector (no intercept)

Survival Analysis 3

Partial Likelihood - Cox Model

$$P(\text{Individual } x_{(i)} \text{ has event at } t_{(i)} \mid \text{someone has event at } t_{(i)}) = \frac{h(t_{(i)} | x_{(i)})}{\sum_{j \in R_i} h(t_{(i)} | x_j)} = \frac{h_0(t_{(i)}) \exp(x_{(i)}^\top \beta)}{\sum_{j \in R_i} h_0(t_{(i)}) \exp(x_j^\top \beta)} = \frac{\exp(x_{(i)}^\top \beta)}{\sum_{j \in R_i} \exp(x_j^\top \beta)}$$

$$\Rightarrow \text{Partial Likelihood: } PL(\beta) = \prod_{i=1}^m \frac{\exp(x_{(i)}^\top \beta)}{\sum_{j \in R_i} \exp(x_j^\top \beta)}$$

$\Rightarrow$ with observed data (t_i, x_i, d_i) , $i=1, \dots, n$: Maximum PL Estimate: $\hat{\beta} = \arg \max_{\beta} (PL(\beta))$

$$\hookrightarrow \text{to solve: } \frac{\partial \log PL(\beta)}{\partial \beta} = \sum_{i=1}^m \left(x_{(i)} - \frac{\sum_{j \in R_i} x_j \exp(x_j^\top \beta)}{\sum_{j \in R_i} \exp(x_j^\top \beta)} \right) \Rightarrow \hat{\beta} \approx \mathcal{N}(\beta, \mathcal{I}^{-1}(\hat{\beta}))$$

Test for Linear Hypotheses

$$H_0: \mathbf{A}\beta = \mathbf{c} \text{ vs. } H_1: \mathbf{A}\beta \neq \mathbf{c}$$

- **Wald statistic:** $W = (\mathbf{A}\hat{\beta} - \mathbf{c})^\top (\mathbf{A}\hat{V}(\hat{\beta})\mathbf{A}^\top)^{-1} (\mathbf{A}\hat{\beta} - \mathbf{c})$
- **(Partial) log-likelihood-ratio statistic:**
 $lq = 2 \cdot (\log PL(\hat{\beta}) - \log PL(\tilde{\beta}))$ with $\tilde{\beta}$ PL estimate under H_0 .

Under H_0 , both $\overset{a}{\sim} \chi^2(\text{rg}(\mathbf{A}))$.

Estimation of Survival Function

Breslow estimate for H_0 :

$$\hat{H}_0(t) = \sum_{t_{(i)} \leq t} \hat{h}_{0i} = \sum_{t_{(i)} \leq t} \frac{1}{\sum_{j \in R_i} \exp(x_j^\top \hat{\beta})}$$

$$\hat{S}_0(t) = \exp(-\hat{H}_0(t))$$

$$\hat{S}(t|x) = \hat{S}_0(t)^{\exp(x^\top \hat{\beta})}$$

For the general non-discrete case, the estimator of h_{0i} is

$$\hat{h}_{0i} = \frac{1}{\sum_{j \in R_i} \exp(x_j^\top \hat{\beta})} \cdot \frac{1}{(t_{(i)} - t_{(i-1)})}$$

Time-dependent covariates

$$h(t, x(t)) = h_0(t) \cdot \exp(x^\top(t) \cdot \beta) \Rightarrow \text{not proportional anymore } \frac{h(t, x_1(t))}{h(t, x_2(t))} = \exp((x_1(t) - x_2(t)) \cdot \beta)$$

Time-dependent effects

$$h(t, x) = h_0(t) \cdot \exp(x^\top \cdot \beta(t)) \Rightarrow \frac{h(t, x)}{h(t, x_{ref})} = \exp(\beta(t)) \Rightarrow \text{function of time}$$

Survival Analysis 4

Piecewise exponential model

$$h(t|\mathbf{x}_i) = h_{0j} \exp(\mathbf{x}_i^\top \boldsymbol{\beta}) = \exp(\beta_{0j} + \mathbf{x}_i^\top \boldsymbol{\beta}) = \exp(\eta_{ij}) =: h_{ij} \quad \forall t \in I_j$$

$$\Rightarrow \text{Likelihood: } L(\boldsymbol{\beta}, \boldsymbol{\beta}_0) = \prod_{i=1}^n h(t_i|\mathbf{x}_i)^{d_i} \exp\left(-\int_0^{t_i} h(s|\mathbf{x}_i) ds\right)$$

where $\boldsymbol{\beta}_0 = (\beta_{01}, \dots, \beta_{0J})$.

Since $h(s|\mathbf{x}_i)$ is piecewise constant, the integral in the log-likelihood can be easily solved.

We define pseudo-data

- event in interval I_j : $d_{ij} = \begin{cases} 1 & t_i \in I_j \text{ and } d_i = 1 \\ 0 & \text{else} \end{cases}$
- time at risk in interval I_j : $t_{ij} = \begin{cases} a_j - a_{j-1} & a_j < t_i \\ t_i - a_{j-1} & a_{j-1} < t_i \leq a_j \\ 0 & t_i \leq a_{j-1} \end{cases}$
- "offset": $o_{ij} = \log t_{ij}$ ($o_{ij} = -\infty$ for $t_{ij} = 0$)

Data in "standard" time-to-event format

ID (i)	t(i)	status(i)	age(i)
1	1.3	0	31
2	0.5	0	67
3	2.7	1	42

→ transform to PED using
 $a_0 = 0, a_1 = 1, a_2 = 1.5, a_3 = 3$

Data in PED format

ID (i)	j	interval	status(i,j)	t(i,j)	t(j)	age(i)
1	1	(0,1]	0	1	1	31
1	2	(1,1.5]	0	0.3	1.5	31
2	1	(0,1]	0	0.5	1	67
3	1	(0,1]	0	1	1	42
3	2	(1,1.5]	0	0.5	1.5	42
3	3	(1.5,3]	1	1.2	3	42

$t_j = a_j$

PEM Likelihood

- Assuming $d_{ij} \stackrel{iid}{\sim} Po(\mu_{ij} = h_{ij} t_{ij})$, we can show that the resulting Poisson likelihood is proportional to a proportional hazards model with piecewise constant hazards.
*Poisson (piecewise const. hazard rate) * (time at risk)*
- Once the data is transformed, we can apply Poisson regression models to obtain hazard estimates for right-censored (and left-truncated) data (Holford, 1980; Laird et al., 1981; Friedman, 1982)
- Setting $\kappa_j = t(j)$ yield estimates identical to the Cox model
so we can fit Cox model with poisson regression
- In this framework "time" is just another covariate
- A smooth function of time $f(t)$ (e.g. penalized splines) can be interpreted as the baseline hazard. Interactions of time with other covariates yields time-varying effects.
- Because the data is split into intervals, time-dependent covariates can be easily integrated by updating their value in each interval.

Trade-off: Number of cut points J

- small J : *is the parameter that can be tuned*
 - crude approximation of the baseline hazard
 - low computational cost
 - stable
- large J :
 - more flexible approximation
 - high computational cost
 - unstable

